Adaptive Blood Drive Scheduling

A Multi-Armed Bandit Approach to Reduce Blood Type Shortages

Diya Patel (J043), Muskaan Sheth (J059), Aaria Srivastav (J066)

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Abstract. Blood banks often run short of specific blood types, O-negative in particular, because the weekly choice of where and when to hold a donation drive follows fixed rotation and habit rather than evidence. We model this choice as a *non-stationary multi-armed bandit* with a persistent blood-stock state: each location and time-slot is an arm, and the reward is the number of units collected, weighted by how depleted each blood type currently is. Turnout parameters are calibrated from the UCI Blood Transfusion Service Center dataset (748 donors), and the problem scale follows India's e-RaktKosh / NBTC data. Over a 104-week simulation averaged across 10 seeds, Discounted UCB and Thompson Sampling collected about **54% more** shortage-weighted blood than fixed rotation and about **84-91% more O-negative units**, while a greedy policy gained only 17%. A sensitivity study shows the gain depends on discounting old observations.

1. Introduction and Problem Statement

1.1 Background and motivation

Blood cannot be manufactured, red cells expire in about six weeks and platelets in under a week, so supply depends on a steady flow of voluntary donors. A blood bank can only influence that flow through one lever: where and when it runs donation drives. Most banks rotate through a familiar list of sites because no system exists to compare site performance. The consequence is a persistent shortage of particular types even when aggregate volume looks healthy. O-negative is the universal-donor type and is in constant clinical demand, yet only about 7% of the population has it, so a schedule that is blind to blood type under-collects exactly the units hospitals need most.

1.2 Problem statement

Given $K = 8$ candidate drive configurations (arms) and a horizon of $T = 104$ weeks, choose an arm a_t each week so as to maximise the cumulative shortage-weighted reward $R = \Sigma\, r_t$. Three properties make this a reinforcement learning problem rather than a forecasting problem: (i) **bandit feedback**, the bank observes the outcome only of the arm it actually chose; (ii) **non-stationarity**, true turnout drifts with academic calendars, corporate campaigns and festivals; and (iii) **persistent state**, the blood-stock vector carries over from week to week, so today's collection changes tomorrow's reward.

1.3 Objectives

1. Formulate blood-drive scheduling as a non-stationary bandit with a stock-dependent reward (a lightweight MDP).
2. Implement and compare two practice-like baselines (Fixed Rotation, Greedy) and the RL methods UCB1, Discounted UCB and Thompson Sampling.
3. Quantify gains in shortage-weighted reward and O-negative units, and explain them through exploration behaviour and the effect of discounting.

1.4 Dataset

The study uses three public sources, each with a distinct role (Table 1). The simulator's behavioural parameters are *calibrated* from the first source; the other two fix the scale and the reward weighting. No public dataset records turnout per drive location, which is why a calibrated simulator, rather than offline logged data, is used for evaluation.

Table 1. Data sources and their role in the project.

Dataset	Content	Role
UCI Blood Transfusion Service Center (Yeh, 2008; CC BY 4.0)	748 real donor records from Hsin-Chu City, Taiwan; 5 attributes: Recency, Frequency, Monetary (c.c.), Time, and Donated (binary target: gave again in the observed period)	**Calibration:** return rate by loyalty (frequency quartile) is 14.4% for the lowest tier and 36.6% for the highest; these anchor the base turnout probability of every arm
India e-RaktKosh / NBTC (Harvard Dataverse extract)	35 states/UTs (309 variables), 616 districts (39 variables), 5,685 facilities; volume, accreditation, wastage, ownership, accessibility	**Scale and context:** realistic structure of the problem; future calibration of regional and seasonal effects
Population blood-type distribution (Red Cross / WHO)	Share of each ABO/Rh type (O-negative about 7%)	**Reward weighting:** type mix of collected blood and priority of scarce types

The UCI table also shows why loyalty matters for scheduling: a site whose visitors are mostly repeat donors converts far better than one dominated by first-timers, and the simulator encodes this through four turnout tiers (14.4% to 36.6%). Only the two end-points come from the data; the two middle tiers (21.8%, 29.2%) are linear interpolations, an assumption we state explicitly.

2. Literature Review

Bandits and the exploration-exploitation trade-off. The bandit problem was posed by Thompson (1933) in the context of clinical trials and formalised as sequential experimental design by Robbins (1952). Sutton and Barto (2018) treat it as the simplest reinforcement learning setting: an agent estimates action values Q(a) from sampled rewards, using the incremental update $Q_{n+1} = Q_n + (1/n)(R_n - Q_n)$, and must balance exploiting the best estimate against exploring alternatives. Auer, Cesa-Bianchi and Fischer (2002) introduced UCB1, which adds a confidence bonus to each estimate and achieves logarithmic regret without any random-exploration parameter.

Non-stationary bandits. Sample-average estimates and plain UCB assume fixed reward distributions and therefore fail when the best arm changes. Sutton and Barto recommend a constant step size so that recent rewards dominate. Garivier and Moulines (2011) analysed two UCB variants for switching environments, Discounted UCB (exponentially down-weighted history) and Sliding-Window UCB, while Besbes, Gur and Zeevi (2014) characterised the achievable regret under a bounded total variation of the rewards.

Bayesian bandits. Thompson Sampling keeps a posterior over each arm's mean reward, samples from it, and plays the arm with the highest sample. Agrawal and Goyal (2012) proved logarithmic regret bounds, Chapelle and Li (2011) showed it is empirically competitive with or better than UCB, and Russo et al. (2018) give a unifying tutorial. Its exploration fades naturally as posteriors sharpen, and discounting counts restores it under drift.

Applications and blood supply. Li et al. (2010) used bandits for personalised news recommendation, a large-scale example of choosing among options when feedback is partial. In blood logistics, the review by Beliën and Force (2012) shows that most work addresses inventory, production and routing with forecasting or mathematical programming; Yeh et al. (2009) modelled donor return behaviour with a recency-frequency-monetary (RFM) formulation, the origin of the dataset used here. We found no work that treats the choice of *drive location and time* as an online learning problem under drift.

Regret and evaluation. Bandit algorithms are judged by regret, the gap between the reward of the best arm and that of the arm actually played, summed over time. UCB-type and Thompson methods achieve logarithmic regret when arm means are fixed, but any method that stops exploring suffers linear regret once the best arm moves. This motivates the discounted variants and the long, drifting horizon used in our experiments, where we measure cumulative reward directly because the true per-week optimum depends on the evolving stock.

From bandits to MDPs. A bandit has no state: actions do not change future rewards. Our reward depends on the stock vector, which the previous drives altered, so the problem sits between a bandit and a Markov Decision Process, where the Bellman equation $V(s) = \max_a E[\, r + \gamma V(s') \,]$ defines optimal behaviour and methods such as Q-learning (Watkins and Dayan, 1992) learn it from experience. We use the bandit view for the decision and keep the MDP as the extension path (Section 5).

Table 2. Methods compared in this project.

Method	Handles drift?	Exploration mechanism	Role
Fixed Rotation	No (ignores data)	None, cycles through arms	Current practice
Greedy	No	Pure exploitation of historical mean	Naive data-driven
UCB1	No	Optimism: confidence bonus	Stationary RL baseline
Discounted UCB	Yes (gamma)	Optimism on discounted counts	Proposed
Thompson Sampling	Yes (gamma)	Posterior sampling, discounted	Proposed

Research gap. Existing work either optimises downstream blood logistics or studies bandits in other domains. This project bridges them: a non-stationary bandit with a stock-dependent, type-weighted reward for an operational decision made weekly by real blood banks.

3. Methodology

3.1 Overall workflow

Figure 1 shows the pipeline. Real data calibrate a simulated environment; policies interact with it in a closed loop; an experiment runner repeats this across seeds and aggregates the results.

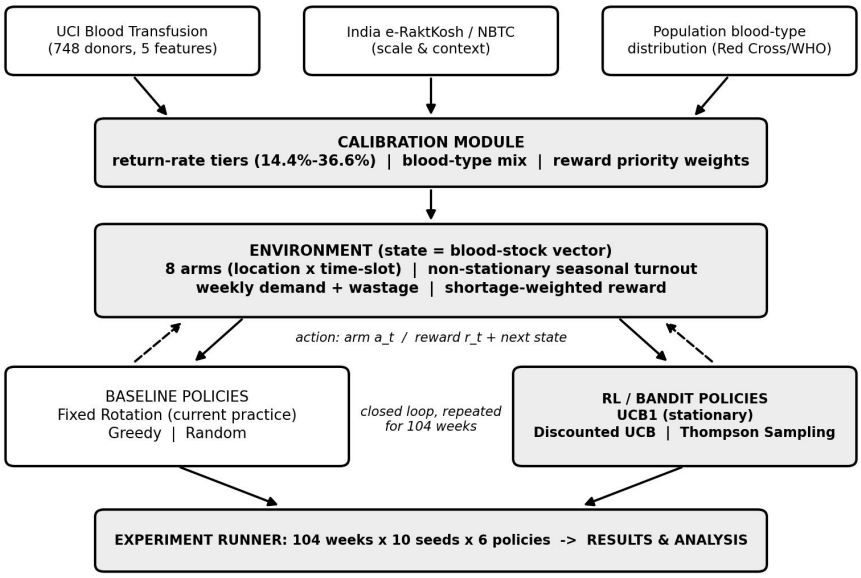


Figure 1. End-to-end system workflow: calibration, environment, policy layer, experiment runner, results.

3.2 Reinforcement learning formulation

Table 3. Mapping of the scheduling problem onto RL elements.

RL element	In this project	Course concept
Agent	Blood-bank scheduler (the policy)	Decision maker
Action a_t	Choose one of 8 location x time-slot arms	n-armed bandit
State S_t	Stock vector over the 8 blood types, as a fraction of target	MDP state
Reward r_t	Units collected x type priority x (0.1 + shortage)	Reward signal
Environment	Seasonal donor turnout, blood-type mix, demand and wastage	Non-stationary dynamics
Learning	Incremental, discounted estimate of Q(a) from observed rewards	Action-value methods

3.3 Simulated environment

Each arm a has a donor pool N_a, a loyalty tier τ_a taken from the UCI-derived tiers, a seasonal multiplier $m_a(t)$ and a fixed blood-type mix π_a (a perturbation of the population distribution, with a higher O-negative share at three sites). In week t, pulling arm a generates:

$$p(a,t) = min(0.95, \tau_a \cdot m_a(t)), \quad n \sim Binomial(N_a, p(a,t)), \quad u \sim Multinomial(n, \pi_a)$$

$$S_{t+1} = max(0, (1 - \omega) \cdot S_t + u - d_t), \quad d_t \sim Poisson(D \cdot \delta), \quad \omega = 2\%, \; D = 90 \; units/week$$

Here u is the vector of units collected per blood type, d_t is hospital demand (δ gives each type's demand share, with O-negative over-represented because it is the universal donor), and ω is wastage. The target stock S* is two weeks of demand and the per-type shortage is $s_k = clip(1 - S_k / S^*_k, 0, 1)$. Table 4 lists the arms; their seasonality creates the non-stationarity: campus turnout collapses in vacations, corporate drives peak around campaign months, malls peak in the festival season, the community centre slowly declines, and a new Tech-Park site launches in week 40 and ramps up.

Table 4. The eight arms (location x time-slot) and their drift pattern.

Arm	Tier	Pool	Mean donors	Non-stationary pattern
Campus Weekday	1	300	34	Low in summer/winter break (x0.25-0.35), exam-period peaks (x1.35)
Campus Weekend	1	110	15	Weak academic dip in summer (x0.6)
Corporate Lunch	3	170	52	CSR campaign peaks in two windows per year (x1.9), x0.75 otherwise
Corporate Weekend	2	70	17	Flat
Community Centre	4	130	41	Flat until week 30, then slow decay to x0.55

Arm	Tier	Pool	Mean donors	Non-stationary pattern
Mall Weekend	1	250	39	Festival-season peak (x1.8), x0.85 otherwise
Hospital Camp	4	60	24	Flat
Tech-Park Evening	3	90	45	New site: x0.3 until week 40, then ramps up to x2.6

3.4 Reward design

Raw volume is the wrong objective, because 100 units of a plentiful type are worth less to a hospital than 100 units of a depleted one. The reward is therefore

$$r_t = \Sigma_k \, u_{t,k} \cdot c_k \cdot (0.1 + s_{t,k})$$

where c_k is a clinical priority weight (O-negative 2.0; A-negative 1.3; B-negative and AB-negative 1.5; all others 1.0). The 0.1 floor keeps a small value on units of well-stocked types. Rewards are divided by 100 before being given to the bandit so that confidence bonuses are on a comparable scale. Because $s_{t,k}$ depends on stock, the reward is state-dependent, which is what places the problem between a bandit and an MDP.

3.5 Policies

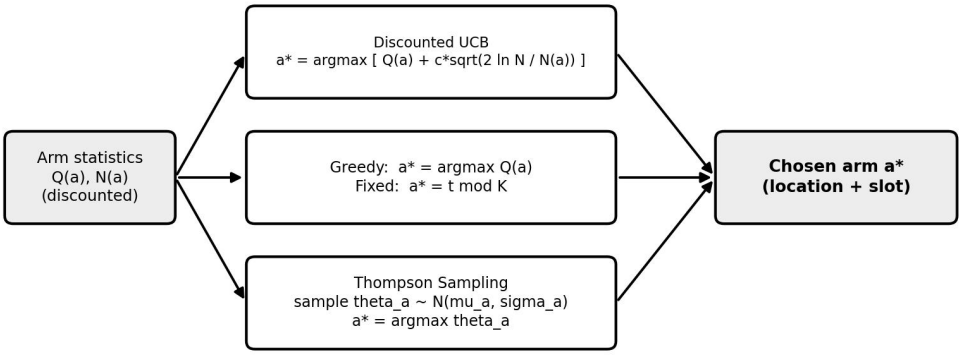


Figure 2. Decision logic of each policy: all read the same arm statistics and differ in how they pick an arm.

Fixed Rotation plays a_t = t mod K and **Random** plays uniformly; **Greedy** plays each arm once and then always the arm with the highest all-time mean. **UCB1** selects a* = argmax [Q(a) + c·√(2 ln N / N(a))]. **Discounted UCB** uses the same rule but replaces counts and reward sums by discounted versions, $N_t(a) = \Sigma_s \, \gamma^{t-s} \cdot 1[a_s = a]$ and $Q_t(a) = \Sigma_s \, \gamma^{t-s} \cdot r_s \cdot 1[a_s = a] \, / \, N_t(a)$, so old evidence fades. **Thompson Sampling** keeps a Gaussian posterior per arm, $\mu_a = (S_a + 0.5)/(N_a + 1)$ and $\sigma_a = \sigma_0/\sqrt{(N_a + 1)}$, with S_a and N_a discounted by γ, samples $\theta_a \sim N(\mu_a, \sigma_a^2)$ and plays argmax θ_a. All learners update incrementally after each week.

3.6 Weekly interaction loop and experimental protocol

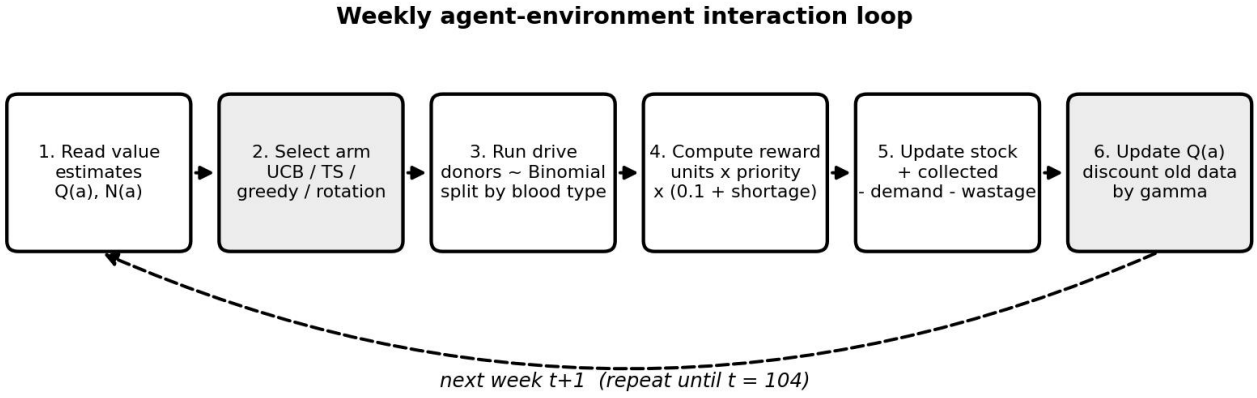


Figure 3. One simulated week: select, run the drive, compute the reward, update stock, update the estimates.

Every policy is run for T = 104 weeks on 10 random seeds (0-9), with the same seeds for all policies. Hyperparameters were set by a small sweep (c from 0.05 to 0.3, γ from 0.90 to 0.98) and are listed in Table 5. We report cumulative reward, percentage improvement over Fixed Rotation, O-negative units, total units, arm-selection shares, and a sensitivity study over γ.

Table 5. Experimental settings.

Setting	Value	Setting	Value
Horizon / seeds	104 weeks / 10	UCB bonus c	0.1

Setting	Value	Setting	Value
Weekly demand D	90 units	Discount factor γ	0.95
Wastage ω / target	2% / 2 weeks	TS prior / σ_0	mean 0.5, strength 1 / 0.1
Initial stock	60% of target	Reward scale	÷ 100

4. Results

Table 6 summarises the 10-seed averages. Both adaptive RL policies clearly outperform Fixed Rotation and Greedy, and they are statistically indistinguishable from each other.

Table 6. Results over 104 weeks, mean of 10 seeds (± standard deviation across seeds).

Policy	Cumulative reward	vs Fixed	Range across seeds	O-neg units	Total units
Fixed Rotation	3783 ± 58	0%	-	264	3132
Random	3933 ± 299	+4.0%	-6.0% to +16.8%	272	3302
Greedy	4438 ± 191	+17.3%	+12.3% to +32.6%	381	3687
UCB1 (stationary)	5085 ± 222	+34.4%	+23.9% to +42.4%	352	4486
Discounted UCB	**5835 ± 187**	**+54.2%**	+45.6% to +61.8%	485	5101
Thompson Sampling	**5813 ± 296**	**+53.7%**	+44.1% to +66.6%	503	5112

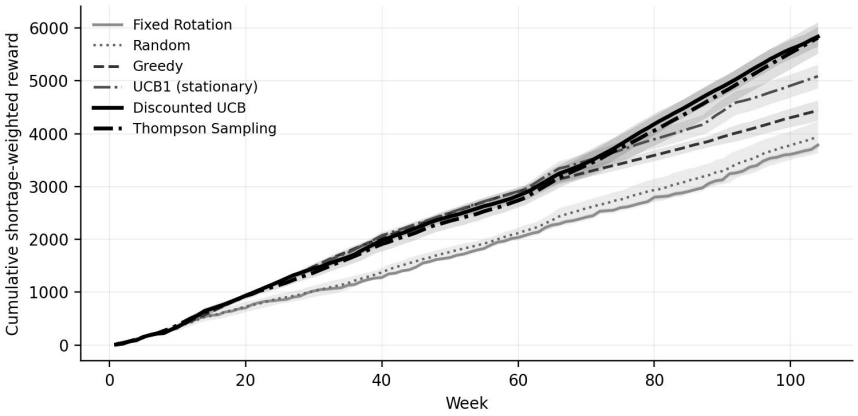


Figure 4. Cumulative shortage-weighted reward (mean of 10 seeds; shaded band = ±1 standard deviation).

Discounted UCB and Thompson Sampling beat Fixed Rotation on every one of the 10 seeds (minimum gain +44%). Relative to Fixed Rotation they collected about 63% more units overall and 84% (D-UCB) and 91% (TS) more O-negative units. Greedy collected 44% more O-negative units but only 17% more reward, and UCB1 sits in between.

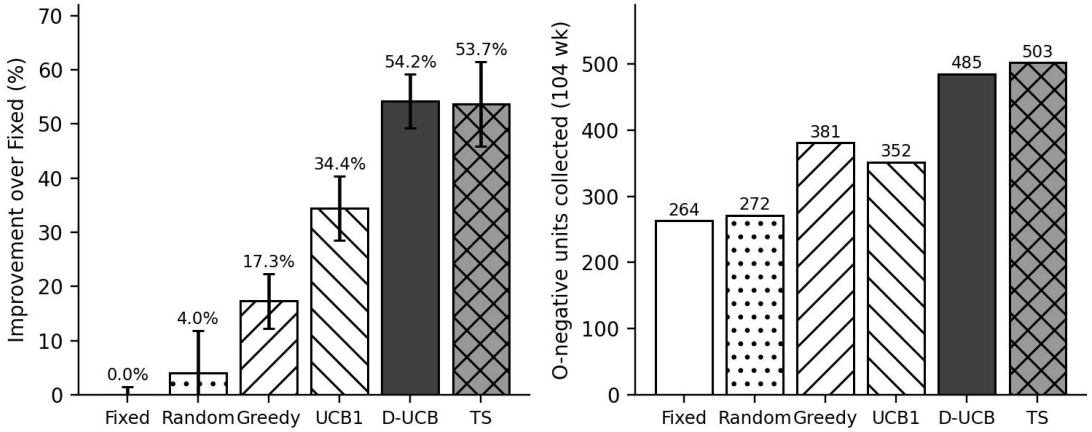


Figure 5. Left: improvement over Fixed Rotation (error bars: ±1 s.d.). Right: O-negative units collected.

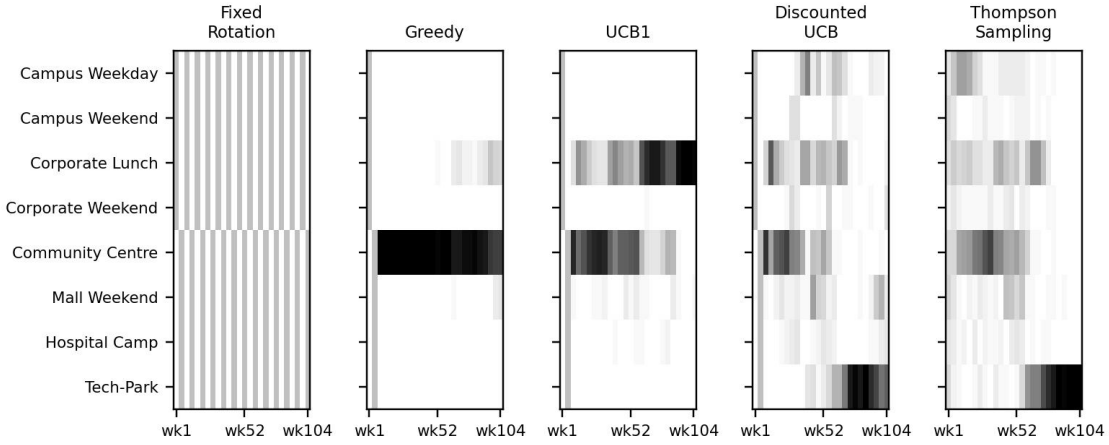


Figure 6. Arm-selection frequency over time (darker = chosen more often; each column = 4 weeks, averaged over seeds).

5. Analysis

Why adaptive policies win. Fixed Rotation gives every arm 12.5% of the weeks, so it earns the *average* arm's reward, whereas the weeks' best arm yields roughly twice that. Any policy that learns which arms are good gains over it; the question is how long it keeps being right.

Greedy locks in. Figure 6 shows Greedy committing to the Community Centre from about week 5 and staying there for 73-100% of each quarter. The centre was genuinely the best early on, but as it decays and Tech-Park and Corporate Lunch overtake it, Greedy has no mechanism to notice: its all-time average is dominated by old data and it never samples the alternatives. Its curve in Figure 4 bends away from the adaptive policies after week 60.

Stationary UCB1 explores, but forgets too slowly. UCB1 does better than Greedy because its confidence bonus keeps testing other arms, and it correctly finds Corporate Lunch. But its estimates for Tech-Park are built from early, low-turnout weeks, and with undiscounted counts the bonus is too small to overturn them: UCB1 plays Tech-Park in only about 1% of the final-year weeks, while Discounted UCB and Thompson Sampling play it 78% and 97% of the final quarter. This is the clearest evidence that non-stationarity, not just exploration, is the central difficulty.

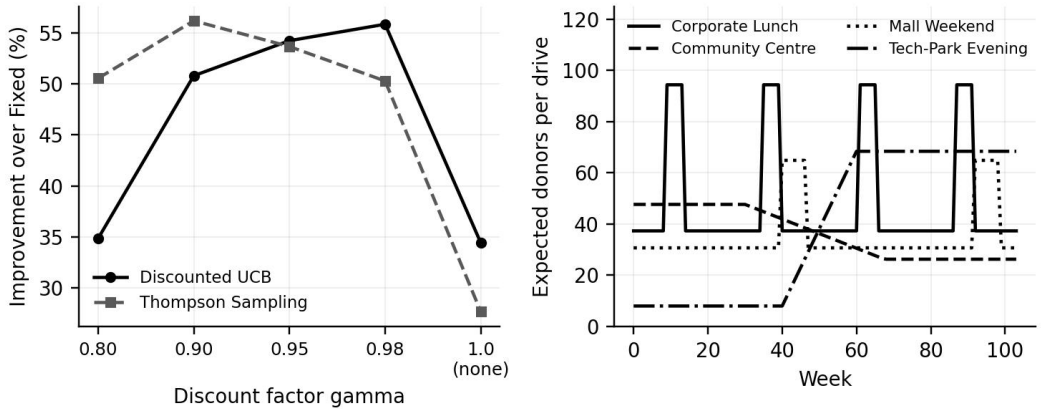


Figure 7. Left: sensitivity to the discount factor γ (γ = 1 means no discounting). Right: expected donors per drive for four arms, showing the drift the policies must track.

Effect of discounting. At γ = 1 Discounted UCB reduces to UCB1 (+34.4%) and Thompson Sampling falls to +27.7%. Introducing discounting lifts both to +50-56% over a broad range of γ (0.90-0.98 for D-UCB; 0.80-0.95 for TS). Too much forgetting (γ = 0.80 for D-UCB) throws away useful data and the gain falls back to +35%. This is the bias-variance trade-off of tracking a moving target: the effective memory length $1/(1-\gamma)$ must roughly match how fast the environment changes.

D-UCB versus Thompson Sampling. Their means differ by under 0.5% and their seed-to-seed spreads overlap, so we cannot rank them. Qualitatively TS commits more sharply to the new best arm (97% vs 78% on Tech-Park in the last quarter) and gives the largest O-negative collection (503 vs 485 units), but it has a wider spread across seeds (±296 vs ±187), consistent with sampling-based exploration being more variable run to run.

Threats to validity. (i) The environment is synthetic: only the two end-point return rates and the blood-type mix are data-derived, while pool sizes and seasonal shapes are assumptions chosen to be plausible. (ii) Demand is set above collection capacity, so stock stays depleted for all policies (O-negative below 25% of target in at least 100 of 104 weeks everywhere); this isolates arm-selection quality but means we cannot claim shortages are *eliminated*, only that

more of the right units are collected. (iii) Hyperparameters were chosen on a small sweep that overlapped the evaluation seeds, so the gains are slightly optimistic; the plateau in Figure 7 suggests they are not fragile. (iv) The absolute percentages depend on how much the arms differ; the ordering Fixed < Greedy < UCB1 < discounted methods is the robust finding.

Extensions. Because the stock carries over, the natural next step is a full MDP: with the state set to the stock vector and a Bellman-based method (e.g. tabular or function-approximation Q-learning) the agent could plan several weeks ahead, for instance deliberately collecting O-negative before a predicted demand spike. A contextual bandit that conditions on the stock vector and calendar features, and calibration against district-level e-RaktKosh data, are further directions.

6. Conclusion

Treating drive scheduling as a non-stationary bandit with a shortage-weighted reward gave a measurable improvement over the status quo: Discounted UCB and Thompson Sampling collected about 54% more weighted blood and about 85-90% more O-negative units than fixed rotation, and about 3x the weighted gain of a greedy policy. The result rests on two ideas from the RL syllabus, optimism or posterior sampling for exploration, and discounting for tracking drift, both simple, interpretable and cheap enough for a blood bank to adopt.

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